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Comparative analysis, Power Quality improvement and Real-Time Simulation of a 3Φ-3W system using DSTATCOM

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***Abstract*—This work describes a 3Φ 3-Level VSC based 6-switch DSTATCOM connected to 3Φ-3W system in shunt at PCC for compensation of linear, non-linear, and unbalanced loads. The working of DSTATCOM is reliant upon SRF technique which is utilized for generating reference currents. Comparative analysis and power quality improvement of the system is achieved in Power Factor Correction (PFC) mode with the extraction of Sending end Power Factor, Receiving end Power Factor, Phase Angle, Magnitude, Mean Power and grid THDs. Simulation models are first designed in MATLAB environment, and then in RT-LAB Software (Real-Time) having Computation and Communication Blocks. A Real-Time Digital Simulator OPAL-RT (4510) is used in which Hardware Synchronized models are embedded and an oscilloscope (RIGOL-MSO5204) with 4-channel capability is used for display of the real-time results.**

Keywords—Distribution Static Compensator (DSTATCOM), Synchronous Reference Frame Theory (SRFT), Point of Common Coupling (PCC), Real-Time Simulation Systems

I. INTRODUCTION

Nowadays distribution systems hold several Power Quality issues particularly due to the delicate equipment usage in residential, industrial and commercial installations. Loads are having new characteristics which result in power quality problems[1-5]. Linear electrical loads usually consume real and apparent power with a power factor of 1 whereas non-linear loads develop harmonics and reactive power problem in addition to the original AC current thus distorting power factor [6]. Besides, any deviation in voltage and current from perfect sinusoidal in terms of magnitude or phase shift is termed as unbalance which is mostly due to distribution side load imperfections [7-11]. These problems are corrected by the DSTATCOM as it is at the top of custom power devices and serves as a retrofit solution to different power quality issues. The main contribution of this paper are as follows:

1) Use of SRFT based control algorithm to control the DSTATCOM connected in shunt at PCC for mitigation of the Power Quality issues that occurred due to the connection of different loads.

2) Comparative analysis of the system, operation and control of DSTATCOM in power factor correction mode involving the extraction of sending end and receiving end power factor, phase angle, magnitude, mean power and grid THDs when different types of Linear, Non-Linear and Unbalanced loads are connected to the Distribution side.

3) An insight about the condition of the power system during working operation when DSTATCOM is connected (to detect the power factor exactly to 1).

II. DESIGN OF DSTATCOM

Fig. 1 represents the schematic design of DSTATCOM.

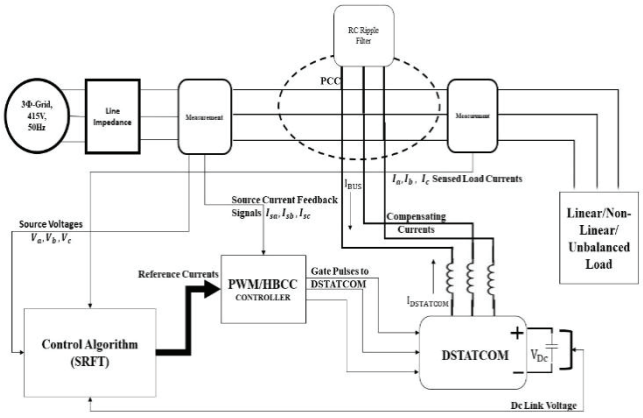


Fig. 1. Schematic Design of 3Φ-3W system with 3Φ 3-Level VSC based 6-switch DSTATCOM at PCC

III. TECHNIQUE FOR CONTROLLING OF DSTATCOM

DSTATCOM is controlled by the SRFT method known as dq0 method. A set of PCC voltages V_a, V_b, V_c and load currents I_a, I_b, I_c are sensed as feedback signals. The sensed current signals are transformed from *abc to dq0* using the control algorithm from digital microprocessor were computation of coordinates of a three-phase (abc) signal takes place in a rotating reference frame (dq0). The angle of the rotating reference frame is given by second input $\theta = \omega t$ using the 3Φ-PLL [15]. The transformation is performed by applying successively an *abc* to *αβ0* and an *αβ0* to *dq0* transformation. It is shown in (1). Then the *dq0* to *abc* transformation is done by the computer microprocessor again which computes a three phase (abc) signal from a space vector in a rotating reference frame (dq0). The angle of the rotating reference frame is given by the second input $\theta = \omega t$ with the help of the same 3Φ-PLL. Then transformation is performed by applying successively a *dq0* to *αβ0* and an *αβ0* to abc^* transformation. At the output we have I_a^*, I_b^*, I_c^* as derived reference currents which are three phase in nature. The transformation is shown in (2)

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$$\begin{bmatrix} I_d \\ I_q \\ I_0 \end{bmatrix} = \frac{2}{3}\begin{bmatrix} cos\theta & sin\theta & 0 \\ -sin\theta & cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & \frac{-1}{2} & \frac{-1}{2} \\ 0 & \frac{\sqrt{3}}{2} & \frac{-\sqrt{3}}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \times \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} \quad (1)$$

$$\begin{bmatrix} I_a^* \\ I_b^* \\ I_c^* \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ \frac{-1}{2} & \frac{\sqrt{3}}{2} & 1 \\ \frac{-1}{2} & \frac{-\sqrt{3}}{2} & 1 \end{bmatrix} \times \begin{bmatrix} cos\theta & -sin\theta & 0 \\ sin\theta & cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} I_d \\ I_q \\ I_0 \end{bmatrix} \quad (2)$$

In the above transformation the compensating action for reactive power, power factor correction and harmonics elimination acknowledges that the source current must give the mean value of direct-axis component of load current along with the active component for maintaining the dc bus and meeting the losses (i_{loss}). Then i_{loss} is given as (3)

$$i_{loss}(k) = i_{loss}(k-1) + \boldsymbol{K_p}[ev_{dc}(k) - ev_{dc}(k-1)] + \boldsymbol{K_i} ev_{dc}(k) T_d \quad (3)$$

where $ev_{dc}(k) = V_{dc}^* - V_{dc}(k)$ is the error between the reference V_{dc}^* and sensed V_{dc} at the Kth sampling instant. K_p and K_i are the proportional and integral gains of the PI controllers. The reference source current is as given by (4)

$$I_D^* = I_{dDC} + i_{loss} \quad (4)$$

The sensed voltages go inside the controller of the control algorithm, and they are as V_a, V_b, V_c respectively. The mathematical block inside the compensator controller computes the amplitude (V_t) of the AC terminal voltages at PCC and is derived from AC source voltages (V_{sa}, V_{sb}, V_{sc}) as given by (5)

$$V_t = \sqrt{\frac{2}{3}} \times \sqrt{(V_{sa}^2 + V_{sb}^2 + V_{sc}^2)} \quad (5)$$

Then we have quadrature axis component, and this component regulates the voltage at PCC. Amplitude of the AC terminal voltage (V_t) at PCC is controlled to its reference voltage (V_t^*) using second PI controller. Similarly, evaluating for i_q is given by (6)

$$i_q = i_q(k-1) + \boldsymbol{K_P}[V_t(k) - V_t(k-1)] + \boldsymbol{K_i} V_t(k) \quad (6)$$

where $V_t(k) = V_t^* - V_t(k)$ represents the error between the V_t^* and the actual V_t terminal voltages of the Kth sampling instant. The reference supply quadrature current is therefore given as (7)

$$I_Q^* = I_{q_{DC}} + i_q \quad (7)$$

IV. Results and Discussion

A. Simulation Results for Balanced Linear Loads in Power Factor Correction (PFC) mode

In this section, comparative analysis on 3Φ-3W power system is carried out using DSTATCOM connected to it at PCC which contributes to the correction of power factor of grid as well as PCC under balanced inductive linear loading conditions. Table 1 shows that simulation is run for different linear (RL) loads and exact value of Power Factor came to unity at a Load of 13000W with positive inductive reactive power of 11000VAR, with simulation run time of 1 or 1.5s. The values which are listed in serial no 6 of the table are selected for accurate finding of the waveform results.

Fig. 2 shows that DSTATCOM maintains constant DC Link voltage after 0.25s as if any change in the DC link voltage results in change in the amount of Active Power exchanged with the source and this voltage is kept close to the reference value (700V) under all disturbances. DSTATCOM compensates reactive power needed by load, so, grid reactive power becomes zero, Fig. 3 shows the zoomed view that grid side voltage and current attain sinusoidal form and linearly follow each other when an RL type of load is connected hence Power factor Correction is achieved with no irrelevant phase shift in grid voltage and current. Fig. 4 exhibits the PCC voltages corrected by DSTATCOM. Fig. 5 shows the injected currents by DSTATCOM which balance the three phase grid currents and supply all reactive power demand and part of active power needed by the load. Fig. 6 shows the load voltage and load current of one phase and it can be seen that although the load current lag behind the load voltage which results in lagging load power factor, the DSTATCOM compensates the load reactive power. Fig. 7 shows grid and load mean (average) powers of one phase respectively.

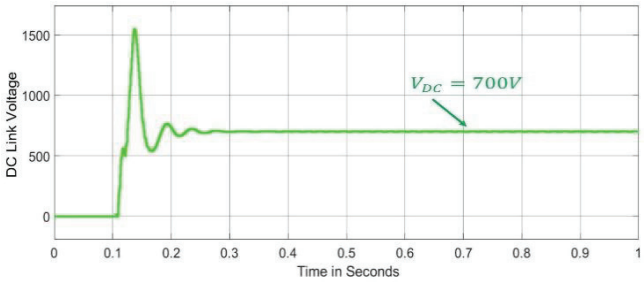


Fig. 2. DC Link Voltage

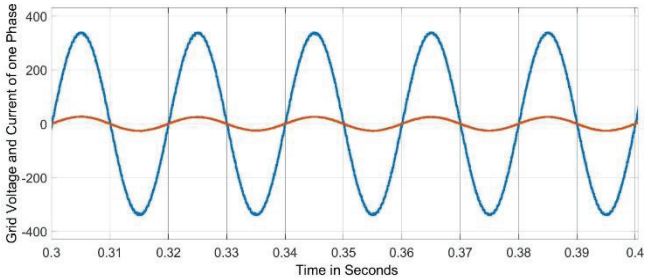


Fig. 3. Grid Voltage and Current of one Phase

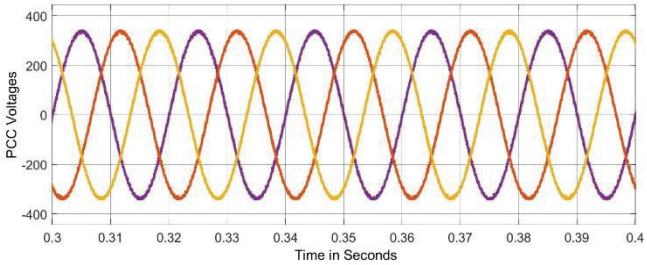


Fig. 4. Corrected PCC Voltages by DSTATCOM

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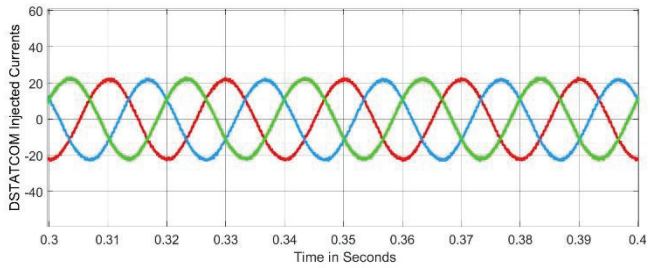


Fig. 5. DSTATCOM Injected Currents

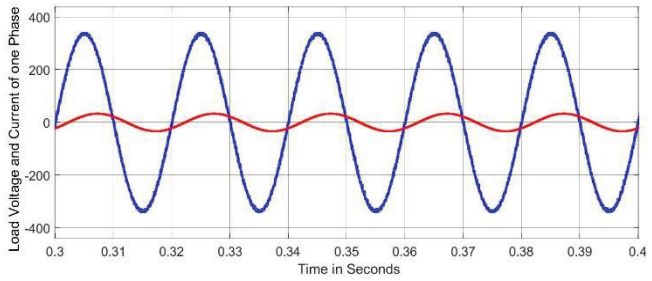


Fig. 6. Load Voltage and Current of one Phase

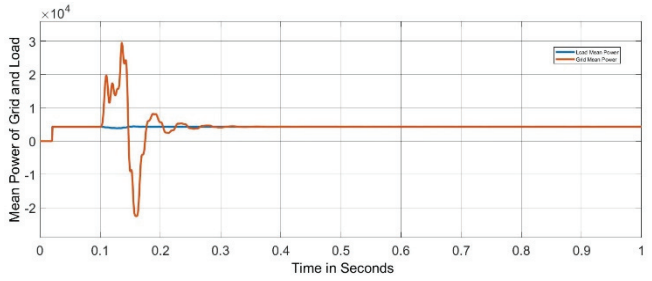


Fig. 7. Mean Powers of Grid and Load

B. Simulation Results of PFC mode and Harmonics Compensation for Non-Linear Loads

In this section, comparative analysis on the same power system is carried out and DSTATCOM is connected to it at PCC. Capability of compensation of DSTATCOM is checked, it corrects the power factor of grid as well as PCC under non-linear loading condition. Non-linear loads have the property that they draw Harmonics which directly effects line currents, in the system line currents will not remain sinusoidal and comprise of fundamental components and a series of harmonic. Harmonics when flow in the entire power network they add-up with fundamental components of grid voltage giving full harmonic distortion. In addition to this, non-linear loads require the reactive power also that must be locally fulfilled and not by the grid. All the source harmonics are cancelled by the DSTATCOM in such a way that it supplies/injects its own harmonics and reactive power required by the 3Φ non-linear load, so that grid supplies only the fundamental part of currents, thus currents become purely sinusoidal and linearly follow the grid voltage giving the unity power factor operation. Table 2 shows the parameters, and that simulation is run for different non-linear loads and exact value of Power Factor came to unity on grid side on connecting a universal-diode-bridge load of 8Ω, 15mH. DSTATCOM is connected at a step time of 0.1s and simulation is run for 1.3s only. It is examined that at this load grid THD is very small as per IEEE-519 Standard, so, these values which are listed in serial no 5 of the table are selected for accurate finding of the waveform results. Fig. 8 shows that inverter DC Link voltage is constant, actually P_{loss} will get converted from AC into DC and it will inject charging current thus maintaining the DC bus voltage constant and this voltage is kept close to the reference value (700V). Fig. 9 shows grid voltage and current of one phase, before compensation there were harmonics present in the system but when DSTATCOM is connected at 0.1s of step time all the odd harmonics are supplied by the active shunt compensator. Fig. 10 shows the grid currents corrected by DSTATCOM which are in sinusoidal form. Fig. 11 shows the DSTATCOM injected three-phase filter currents and these currents will cancel the harmonics at PCC that created by the non-linear load. Fig. 12 depicts the distorted load currents. Fig. 13 show the distorted load voltage and current of one phase. Fig. 14 shows the grid and load mean (average) powers of one phase in case of nonlinear load respectively.

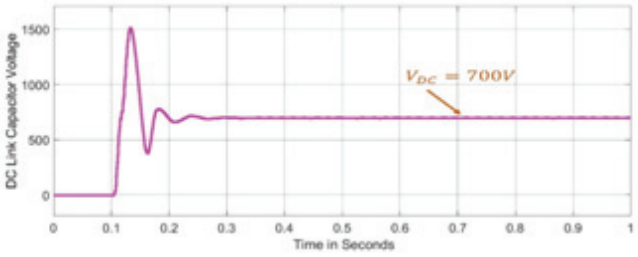


Fig. 8. DC Link Voltage

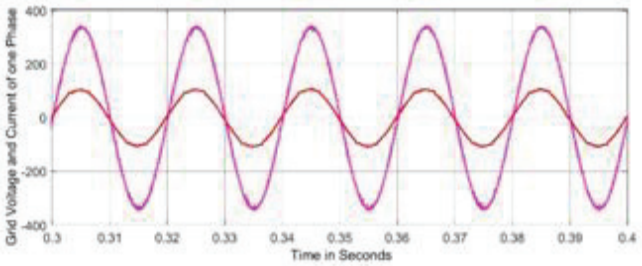


Fig. 9. Grid Voltage and Current of one Phase

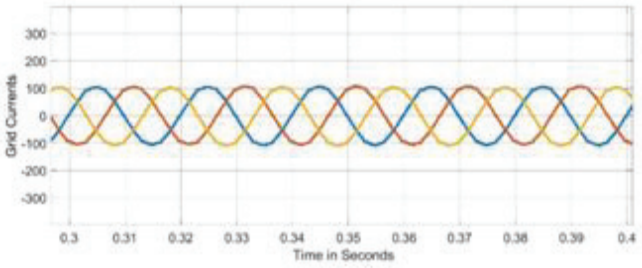


Fig. 10. Sinusoidal Grid Currents

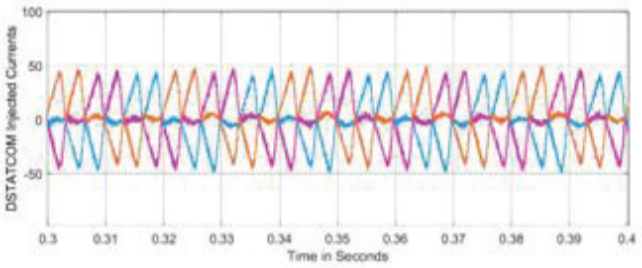


Fig. 11. DSTATCOM Injected Currents

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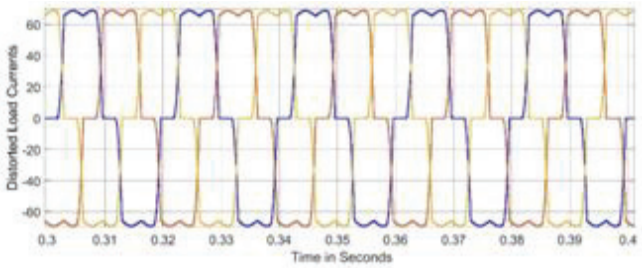


Fig. 12. Distorted Load Currents

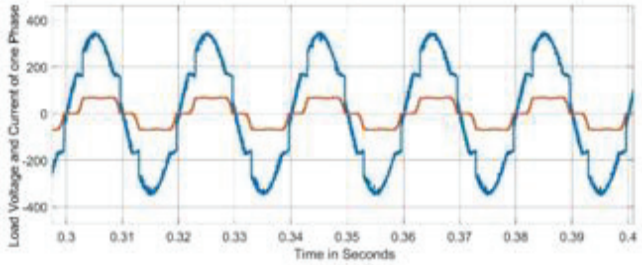


Fig. 13. Load Voltage and Current of one Phase

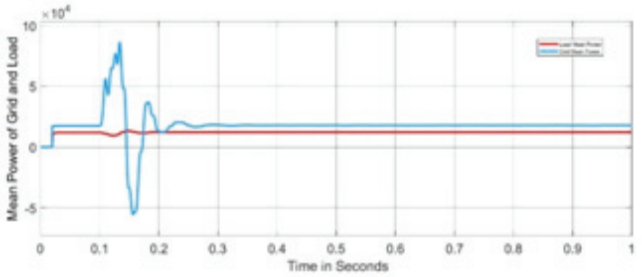


Fig. 14. Mean Powers of Grid and load

C. *Simulation Results of PFC mode for 3Φ Unbalanced RLC Loads*

In this system, comparative analysis is carried out on the same power system and the load connected is unbalanced (RLC) Load. Load currents are unequal as their amplitudes and phase angles are not balanced. The DSTATCOM injects the negative sequence component currents into the load side network so that only positive sequence current should come from the grid side. All the active power comes from the grid side, reactive power from DSTATCOM and grid power factor becomes to unity. Table 3 shows the parameters, and the simulation is run for different unbalanced RLC loads and Power Factor is maintained at 1 on grid side no matter how the system reactive power varies. In the operation the values which are listed in serial no 5 of the table are selected for accurate finding of the waveform results. DSTATCOM when connected makes grid currents in-phase, balanced, sinusoidal and displaced 120^0 apart with respect to supply voltage. At simulation run time of 1.3s it was observed that power factor correction is achieved to unity when a load is put on the three phases in distribution side having effect as a = 10Ω +7mH, b = 5 Ω +10mH, and c = 3Ω+6mH+10μF.

Fig.15 shows the DC link voltage of DSTATCOM which came constant and is maintained at a value of 700V DC. Fig. 16 shows grid voltage and current of one phase, the source current linearly follows the source voltage when DSTATCOM is connected at 0.1seconds. Fig. 17 shows the grid currents corrected by the DSTATCOM after 0.1 seconds. Fig. 18 shows the DSTATCOM injects highly unbalanced currents to compensate for the grid current unbalances. Fig. 19 shows the unbalanced load currents. Fig. 20 shows the grid and load mean (average) powers of one phase respectively.

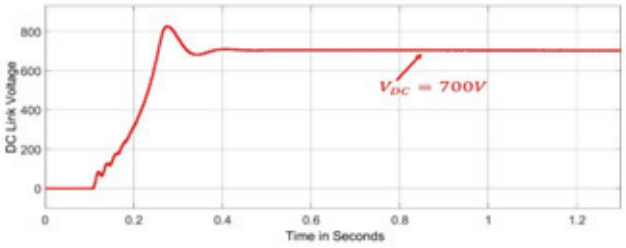


Fig. 15. DC Link Voltage

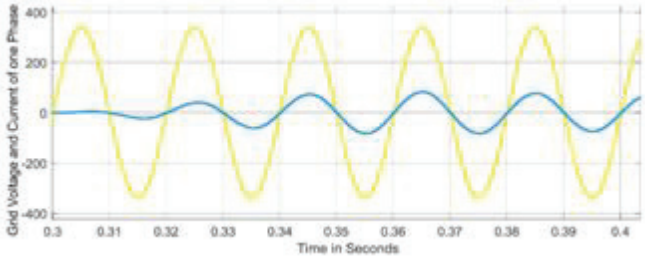


Fig. 16. Grid Voltage and Current of one Phase

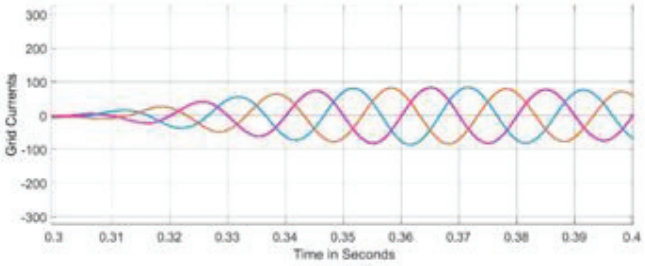


Fig. 17. Grid Currents

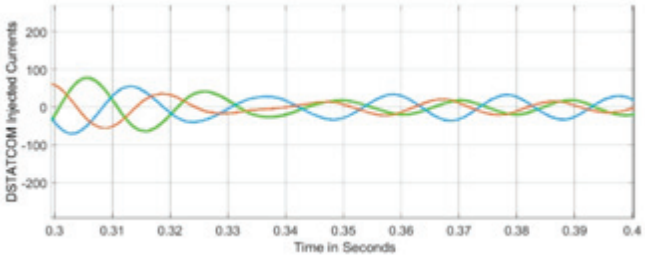


Fig. 18. DSTATCOM Injected Currents

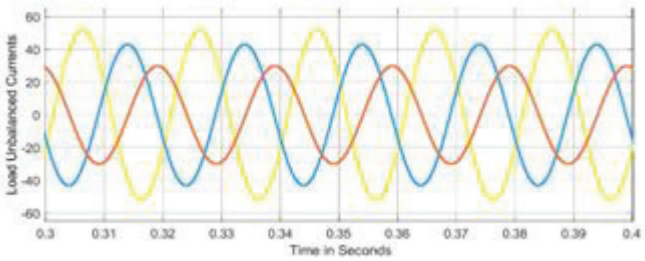


Fig. 19. Unbalanced Currents Due to Load Unbalancing

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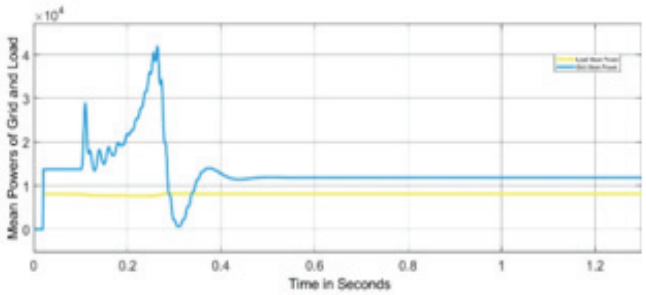


Fig. 20. Mean Power of Grid and Load

D. RT--LAB Real-Time Experimental Results

The Experimental Setup is shown in Fig. 21



Fig. 21. Experimental Setup

1) Real-Time Experimental Results with 3Φ Linear Load in PFC Operation mode with Power Factor 1

Figures 22-24 exhibit the validated Real-Time results of Balanced Linear Inductive Loads in PFC mode.

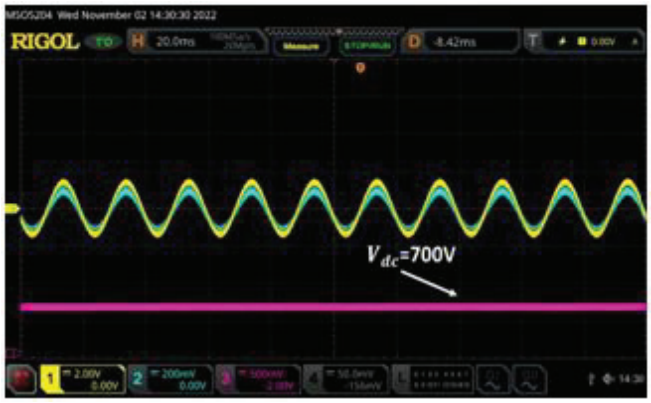


Fig. 22. Exhibits Sinusoidal Grid Voltage and Current of one Phase and Constant DC Link Voltage



Fig. 23. Corrected 3Φ Grid Currents by DSTATCOM (PFC Mode)



Fig. 24. DSTATCOM Injected Currents (PFC Mode)

2) Real-time Experimental Results of PFC Operation mode with 3Φ Non-Linear Load

Figures 25-28 exhibit the validated Real-Time Results of Power Factor Correction mode in case of Non-Linear Load (NLL). Real-Time results are fetched when DSTATCOM is connected at 0.1s of step time and simulation is run in Fixed Step mode. The power factor is exactly achieved to unity on source side at a nonlinear load of 8Ω, 15mH and during this actual operation the Real-Time waveforms were fetched from Oscilloscope. The DSTATCOM injects a signal which suppresses the harmonics content. Grid voltage and current are linearly following each other hence power quality is improved.

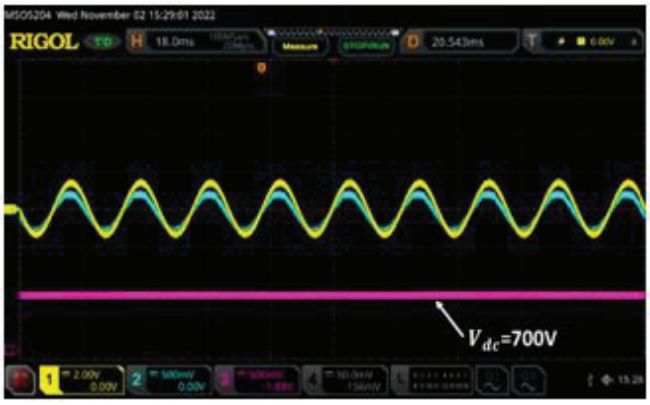


Fig. 25. Exhibits Grid Voltage and Current of one Phase and Constant DC Link Voltage

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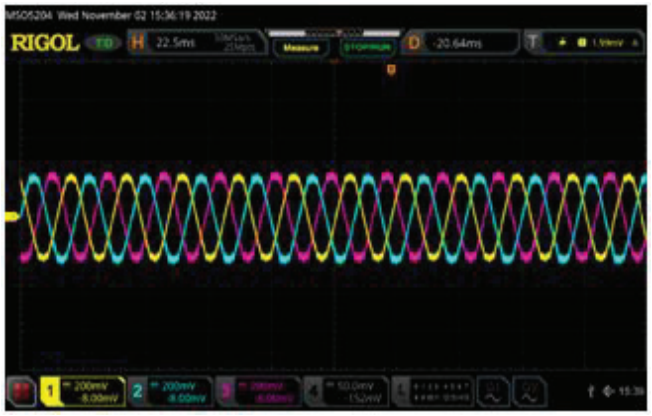


Fig. 26. Corrected 3Φ Grid Currents due to Connection of DSTATCOM at PCC



Fig. 27. Distorted 3Φ Load Currents due to NLL Connection



Fig. 28. DSTATCOM Injected Currents in case of NLL

3) Real-Time Experimental Results of PFC Operation mode with 3Φ unbalanced RLC load

Figures 29-32 exhibit the validated Real-Time Results of Power Factor Correction mode in case of Unbalanced Loads. In this experimental RT-LAB check grid currents were in-phase, balanced and sinusoidal with respect to grid voltage when DSTATCOM was connected in the system at 0.1s of step time. Hence power quality is improved.

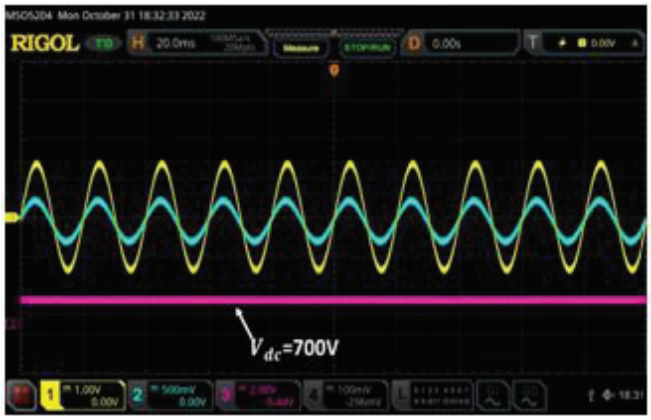


Fig. 29. Exhibits Sinusoidal Grid Voltage and Current of one Phase (PFC Operation) and Constant DC Link Voltage



Fig. 30. Balanced Grid Currents



Fig. 31. The Unbalanced Load Currents because of Unbalanced Connected Load



Fig. 32. Load voltage and current of one phase because the load connected is Linear RLC type

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TABLE I. SIMULATION ON DIFFERENT LINEAR LOADS WITH SYSTEM PARAMETERS FOR PERFECT PFC OPERATION MODE

S No	Time of Simulation Operation	Type of Load Connected (Linear Load)	Sending end Power Factor	Receiving end Power Factor	Grid-Side Phase angle of Current Vector w.r.t Voltage Vector	Magnitude of Grid Current Vector	Grid Mean Power	Load Mean Power
1	0.5 seconds	Active Power = 2000W, Inductive reactive Power = 1500Var	0.9988	0.799	2.81°	3.935	665.7W	666.5W
2	0.7 seconds	Active Power = 4000W, Inductive reactive Power = 3000Var	0.9996	0.7986	1.572°	7.904	1338W	1333W
3	0.9 seconds	Active Power = 8000W, Inductive reactive Power = 6000Var	0.9999	0.7977	0.8996°	15.76	2668W	2664W
4	1.1 seconds	Active Power = 9000W, Inductive reactive Power = 7000Var	0.9999	0.7868	0.7837°	17.7	2997W	2997W
5	1.3 seconds	Active Power = 11000W, Inductive reactive Power = 9000Var	0.9999	0.7709	0.6854°	21.65	3665W	3662W
6	1 or 1.5 seconds	Active Power = 13000W, Inductive reactive Power =11000Var	1	0.7598	0.5701°	25.59	4332W	4326W

TABLE II. SIMULATION OPERATION ON DIFFERENT NONLINEAR LOADS WITH SYSTEM PARAMETERS FOR PERFECT PFC OPERATION MODE

S No	Time of Simulation Operation	Type of Load Connected (Non-Linear Load with RL-Connected)	Sending End Power Factor	Receiving End Power Factor	Grid-Side Phase angle of Current Vector w.r.t Voltage Vector	Magnitude of Grid Current Vector	Grid Mean Power	Load Mean Power	Grid THD, Operation for Number of Cycles 10, Start Time 0.3, Stop Time 0.5
1	0.5 seconds	1Ω, 5mH	0.9999	0.765	-0.6624	417.1	380.54W	348.71W	Grid Current THD =9.46%
2	0.7 seconds	3Ω, 7mH	0.9985	0.909	3.132°	207.3	190.21W	158.77W	Grid Current THD =2.84%
3	0.9 seconds	5Ω, 9mH	0.9984	0.9445	3.283°	145.3	133.54W	102.2W	Grid Current THD =1.52%
4	1.1 seconds	7Ω, 11mH	0.9986	0.9602	3.057°	115.8	106.57W	75.34W	Grid Current THD =1.63%
5	1.3 seconds	8Ω, 15mH	1	0.9654	3.002°	106.3	97.839W	66.609W	Grid Current THD =1.63%
6	1.5 seconds	10Ω, 30mH	0.9988	0.9724	2.823°	92.47	85.336W	9892W	Grid Current THD =2%

TABLE III. SIMULATION OPERATION ON DIFFERENT UNBALANCED LOADS WITH SYSTEM PARAMETERS FOR PERFECT PFC OPERATION MODE

S No	Time of Simulation Operation	Type of Load Connected (Unbalanced RLC Load)	Sending End Power Factor	Receiving End Power Factor	Grid Side Phase angle of Current Vector w.r.t Voltage Vector	Magnitude of Source Current Vector	Source Mean Power	Load Mean Power
1	0.5 seconds	a = 2Ω+1mH, b = 2.5Ω + 7mH, c = 3Ω + 2mH +6μF	0.9612	0.9988	16.01°	143.4	Phase a=126.88W Phase b=113.94W Phase c= 177.66W	Phase a= 104.25W Phase b= 69.66W Phase c= 150.8W
2	0.7 seconds	a = 2.5Ω+ 1.5mH, b = 3Ω+ 6mH, c = 3.5Ω + 2.4mH +6.5μF	0.9812	0.996	11.12°	128.4	Phase a= 115.74W Phase b= 110.45W Phase c= 149.70W	Phase a= 88.77W Phase b= 74.03W Phase c= 119.7W
3	0.9 seconds	a = 3Ω + 4mH, b = 5Ω + 7mH, c = 2.1 Ω+ 3mH +7μF	0.9985	0.9698	3.172°	165.3	Phase a= 151.89W Phase b= 103.29W Phase c= 119.67W	Phase a= 140.42W Phase b= 9596W Phase c= 87.57W
4	1.1 seconds	a = 4Ω + 5mH, b = 6Ω + 9mH, c = 2.5Ω + 4mH +8μF	0.9999	0.9408	0.8436°	135.9	Phase a= 125.13W Phase b= 93.03W Phase c= 97.18W	Phase a= 111.21W Phase b= 7811W Phase c= 67.15W
5	1.3 seconds	a = 10Ω + 7mH, b = 5Ω + 10mH, c = 3Ω + 6mH +10μF	1	0.9814	0.5046°	62.62	Phase a= 57W Phase b= 70.37W Phase c= 70.04W	Phase a= 5523W Phase b= 8180W Phase c= 5460W
6	1.5 seconds	a = 11Ω + 8mH, b = 6Ω + 17mH + 11μF, c = 4Ω + 8mH	0.9951	0.8421	-5.675	79.59	Phase a= 73.05W Phase b= 71.41W Phase c= 58.85W	Phase a= 55.79W Phase b= 5584W Phase c= 4206W

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V. CONCLUSION

DSTATCOM addresses the power quality problems and issues in relation to grid and load. It makes sinusoidal balanced grid currents in case of different connected loads. This work examines the exact operation, simulation and control of DSTATCOM to observe the actual unity power factor (PFC) operation. Execution of Real-Time Experimental Results worked out at the specific loading conditions give the real updated operation of DSTATCOM in the system. The work shall also pave way for designing, operation and maintenance of any upcoming power system.

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